

Cobots and Collaborative Robots: Types and Applications



Introduction

• This article will give an in-depth discussion about collaborative robots.

The article will bring more understanding about:

<https://www.iqsdirectory.com/articles/automation-equipment/collaborative-robots.html>

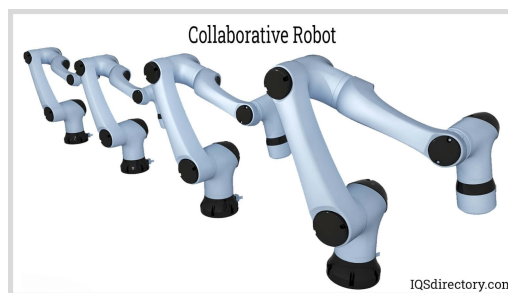
- Applications and Benefits of Collaborative Robots
- And Much More...

Chapter 1: Understanding the Core Principles of Collaborative Robots

In this chapter, we delve into the world of collaborative robots, highlighting crucial factors to consider when choosing the appropriate cobot for your specific applications.

Defining Collaborative Robots

Collaborative robots, commonly known as cobots, symbolize a breakthrough in robotic advancements, redefining automation by operating alongside human workers. Unlike conventional robots, cobots are fitted with sensors that enable environmental awareness. This technology allows them to work safely with humans, transitioning into a protective mode if any disturbance or safety threat is detected, thus minimizing risks associated with traditional robots lacking these safety features.



Automation Equipment

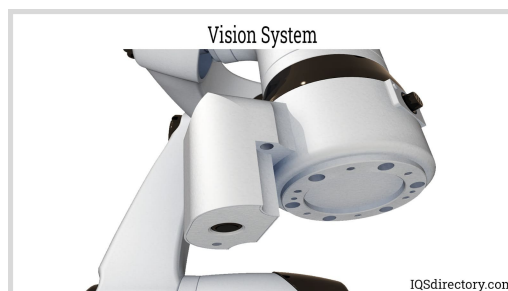
Collaborative robots are a specialized category within automation equipment, a comprehensive range of machinery facilitating automated production processes. This category also includes automation cells, conveyors, industrial robots, and niche equipment like turn-over machines and lifters.

Various Tools for Collaborative Robots

Cobots utilize tools akin to those used by traditional robots, adaptable to diverse positions and applications. Manufacturers worldwide are investing in creating specialized tools for cobots, enhancing their task learning capabilities and versatility. These tools encompass Vision systems, End of Arm Tooling (EOAT), Grippers, Software, Range Extenders, Supply components, and Safety Systems.

Cobot Vision Systems

Vision systems provide collaborative robots with the ability to perceive and decode their environment. Utilizing 3D or 2D cameras, cobots can detect and locate objects, identify patterns, and scan barcodes. This flexibility allows for efficient product handling; unlike traditional systems requiring organized layouts, vision-equipped cobots base actions on visual data. These systems support optical inspections, like detecting product flaws, and their capabilities are further boosted by Artificial Intelligence, offering sophisticated functionalities and improved efficiency.



In the realm of vision systems for collaborative robots, different options exist, tailored to various applications based on adaptability, cost-effectiveness, and ease of operation. Key choices include 3D and 2D camera setups, selected based on the task-specific needs envisioned for the collaborative robot.

2D Vision Systems

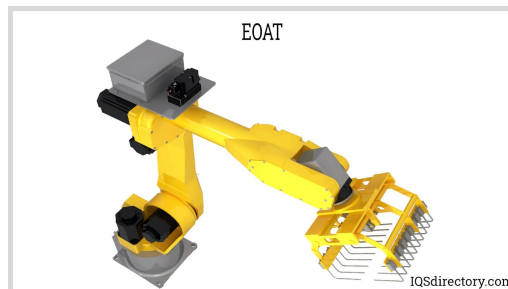
While more economical than 3D cameras, 2D cameras have limited functionalities. They can measure an object's width and length but not its height.

3D Vision Systems

3D cameras offer capabilities to measure an object's width, height, and length, making them highly versatile, although at a higher cost. Their ability to measure height is ideal for dealing with objects of varying elevations or stacked items, proving useful in packing or palletizing operations.

End of Arm Tooling (EOAT)

End of Arm Tooling (EOAT) includes a spectrum of tools attachable to collaborative robots, such as sanders, screwdrivers, sensors, tool changers, glue dispensers, welding apparatus, and soldering tools. Recent advancements in these tools have greatly enhanced their potential, enabling cobots to play a more integral role in amplifying production capacities for various companies.



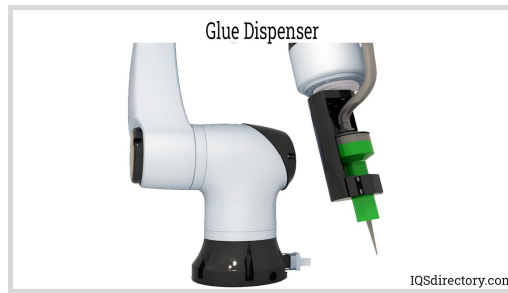
Varieties of EOAT

EOAT presents vast opportunities for training cobots on new tasks. The continuous advancement of these tools will allow cobots to tackle a wider array of applications in the coming future. Below are some examples of the diverse EOAT tools available:

EOAT Glue Dispensers

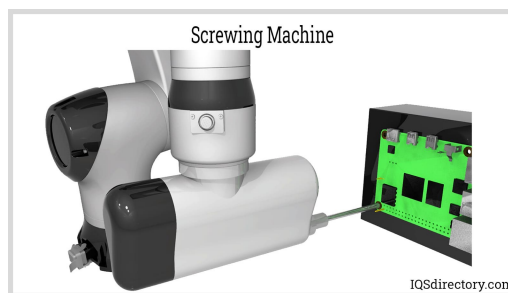
A dispenser is used to affix a glue or sealant syringe to a cobot's end, allowing precise path-following and glue

application with consistency, reducing wastage by precisely controlling glue quantity.



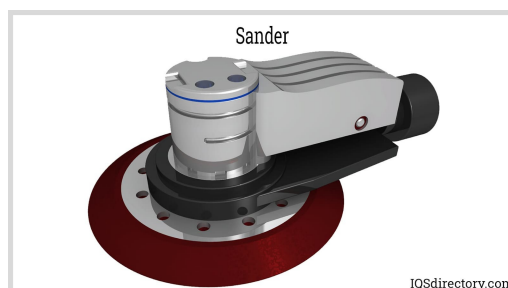
EOAT Screwing Machines

Screwing machines enable cobots to execute fastening tasks independently or alongside human operators. Screwing is monotonous and repetitive, making it ideal for cobots, which can efficiently manage this task over extended durations.



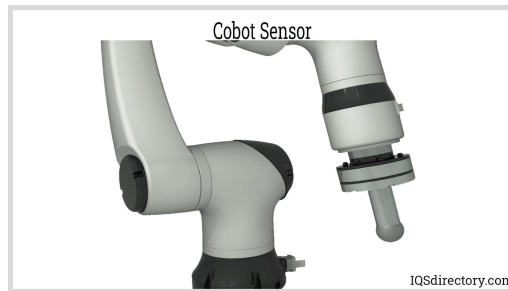
EOAT Sanders

Armed with a sander, a cobot can proficiently sand, polish, or deburr surfaces. Integrated sensors ensure consistent pressure application for uniform results, and user-friendly software allows easy programming for specific paths.



Cobot Sensors

Collaborative robots come with integrated sensors enhancing their capability to work safely with humans. For tasks demanding extra precision, additional sensors, such as an external force torque sensor, can further improve sensitivity.



Cobot Soldering and Welding Tools

Soldering and welding demand a high precision level, making cobots an excellent match due to their accuracy. Equipping a cobot with a soldering iron or welding torch allows for effective automation of these tasks.



Cobot Grippers

Grippers enable collaborative robots to manipulate items by lifting and positioning them in different locations. A variety of grippers exist, including vacuum, magnetic, and finger grippers, designed for specific use cases such as vacuum grippers for boxes or soft grippers for delicate goods. Specifications such as gripping width, force, and payload capacity are crucial for selection.



Different Types of Cobot Grippers

Grippers vary according to material and task type. Selection should consider the weight and material of items to be handled.

Electric Grippers

Electric grippers suit complex tasks with their ability to fine-tune parameters like finger positioning and grip strength.

Pneumatic Grippers

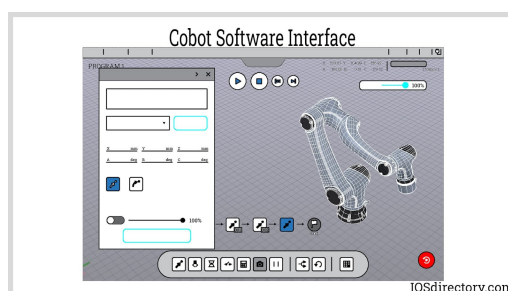
Pneumatic grippers, driven by air, handle simpler tasks. They're lighter, cost-effective, and offer a significant grip force, useful for varied applications.

Finger Grippers

For high-precision work, three- or two-finger grippers are optimal, adapting to objects' shapes to ensure secure handling.

Cobot Software

Numerous resources exist for programming and designing applications for cobots, each robot coming with intuitive software to match its function. Simulation tools aid in development and refinement before implementation.



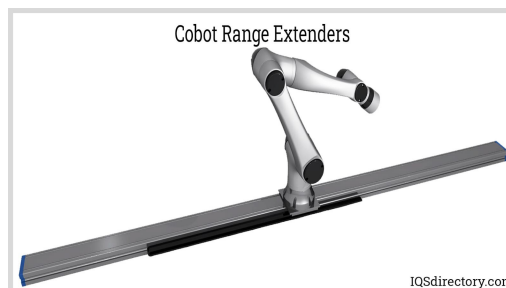
Traditional robots typically require extensive programming time and expertise, generally performing limited predefined actions. Cobots, however, are designed for ease of programming, learning new tasks swiftly with intuitive software and mobile apps. This simplicity allows for quick deployment and the ability to perform a wide variety of tasks.

RoboDK

RoboDK is an affordable, robust robot simulation tool, featuring over 500 robot arms in its library. Its user-friendly interface requires no programming experience, allowing rapid setup for simulations. Supporting offline programming, RoboDK facilitates cobot program development and testing outside production settings, suitable for machine tending, pick- and-place, welding, packaging, and palletizing applications.

Cobot Range Extenders

Range extenders augment cobots' potential by expanding their horizontal and vertical operational reach, thereby enhancing automation capabilities. Typically constrained to their base axis, cobots gain increased mobility with range extenders, which also enhance productivity. For instance, in palletizing, robots can handle higher, more efficient box stacking, and navigate in front of large machinery, executing complex workflows.



Collaborative robots are praised for their ease of application and swift installation. With range extenders, they become even

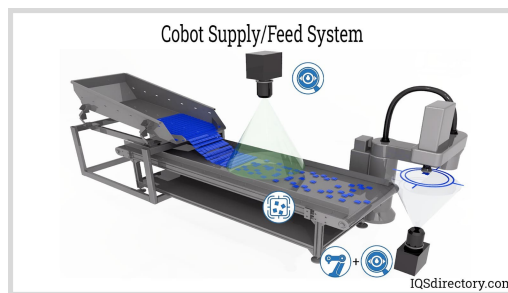
simpler to integrate, aided by user-friendly software for seamless cobot communication setup.

Cobot Safety

Safety measures include tools ensuring safe human-robot interaction, such as barriers and sensors that detect human proximity to the robot.

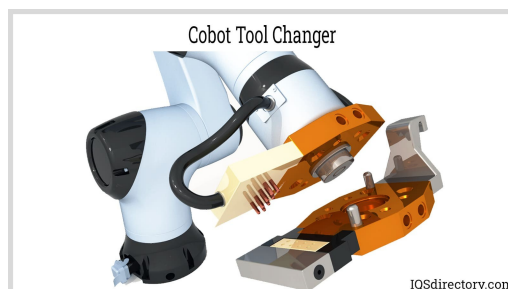
Cobot Supply Systems

Supply systems comprise tools providing cobots with materials for manipulation. For instance, a tool can place screws in specific positions for cobot access, while screw-feeding systems collect and deliver them for assembly processes. Vibrating plates assist in sorting and organizing mixed items.



Cobot Tool Changers

Enhanced cobot versatility arises from various automatic tool changers available on the market. These allow autonomous switching between tools, enabling multiple operation performance. In assembly tasks, a cobot might use a gripper to position components before switching to a screwdriver for final assembly, boosting productivity and return on investment.



Key Factors in Choosing a Collaborative Robot

Several points should be assessed when selecting a collaborative robot:

Collaborative Robot Applications

Cobots are ideally suited for tasks that are perceived as dirty, boring, dangerous, or physically exhausting. Typical uses include assembly, handling materials, inspection, and packaging. These robots are engineered with qualities like reach, precision, and payload capability for efficient human collaboration. While not as fast or powerful as traditional industrial robots, cobots excel in safety and adaptability, filling in workforce gaps and enabling human employees to focus on more strategic tasks.

The ISO 10218-1 and ISO 10218-2 outline four crucial cobot features: stops, hand guiding, speed and separation monitoring, and force and power limitations. These characteristics ensure safe, efficient operation across various applications. Speed and separation monitoring support tasks involving close human-robot interaction, while force and power limits are crucial for delicate operations. These standards facilitate safe, effective cobot integration in diverse settings.

Systems Integrator

Implementing a cobot demands ample expertise. Evaluate whether your team can manage the process or if a systems integrator is needed. Systems integrators offer valuable insights from varied client experiences, providing customized solutions and avoiding the hurdles of starting anew. Their expertise can often ensure cost-effective and smoother cobot integration.

Expert Robotic Integrators

Significant expertise is crucial when integrating a cobot. Determine if your internal team possesses the required skills and time or if you should consult a robotic integrator. These integrators, also called systems integrators, analyze your operations and advise on what would work best based on their extensive experience. They bring cost-effective solutions by preventing the need to create from scratch.

Conducting Risk Assessment

Risk assessment is a vital component when deciding on cobot usage. Despite built-in sensors designed to stop motions and avert injuries, cobots still pose potential risks. Ensuring full safeguarding is essential to secure safe, effective operation for staff protection.

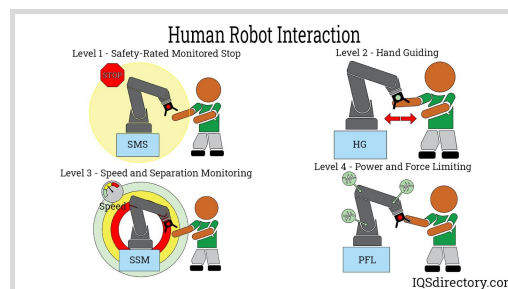
Addressing Collaborative Robot Queries

Open communication is critical for successful cobot implementation. It's essential to engage all involved parties, from top-level executives to employees interacting with the cobot. Ensuring all stakeholders are knowledgeable and aligned can significantly streamline the integration process.

- ◆ What are the benefits of this automation type?
- ◆ What investment levels and ROI projections are expected?
- ◆ How might employee roles change, or can cobots free them for more intellectual tasks?
- ◆ What training and learning opportunities exist, and who manages them? (Systems integrators often include staff training in their packages.)
- ◆ After installation, what support is available? Can
- ◆ cobots help address workforce shortages?
- ◆ What are the company's future automation plans?

Chapter 2: What are the Types of Collaborative Robots?

Collaborative robots, also known as cobots, represent a groundbreaking advancement in industrial robotics and smart manufacturing. These innovative automation solutions are engineered to work safely and efficiently alongside human workers in a shared workspace. The main types of collaborative robots differ in their design, safety mechanisms, and levels of required human-robot interaction, making them adaptable to a wide range of industrial automation and assembly applications. By understanding the differences between these cobot types and the technology driving their development, manufacturers and businesses can make informed decisions on selecting the optimal solution for improved productivity, safety, and return on investment.



Power and Force Limiting Cobots

In most conversations about collaborative robots, Power and Force Limiting (PFL) cobots are the primary focus. These collaborative robots are engineered with advanced safety features, including torque sensors, soft materials, and rounded edges, all contributing to direct human-robot collaboration without the need for traditional safety fences, external light curtains, or complex guarding systems.



To further advance operator safety, these force and power limiting cobots utilize collision detection sensors and safety-rated monitored stop functions that instantly halt operation upon making contact with a human. Due to these robust safety protocols, they are excellent for light payload tasks such as electronic assembly, machine tending, laboratory automation, precision inspection, and packaging in industries demanding flexible automation and human-machine interaction.

Although PFL collaborative robots are typically intended for lighter-duty applications, ongoing advancements in robotics technology are rapidly expanding their capabilities, enabling more diverse use cases across manufacturing, logistics, healthcare, and research environments.

Safety Monitored Stop Cobots

Safety Monitored Stop collaborative robots integrate conventional industrial robot arms with safety monitoring systems such as presence detection devices, interlock gates, and emergency stop sensors. These cobots are ideal for manufacturing processes that require occasional human intervention for quality checks, tool changes, or equipment maintenance.



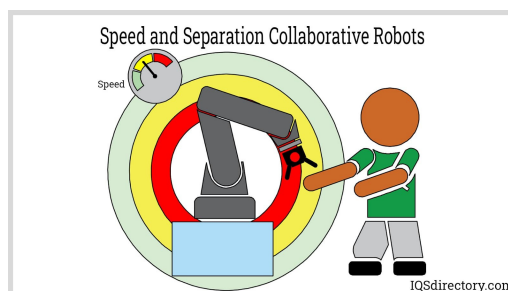
When a human worker enters the robot's workspace, the cobot immediately suspends all motion, ensuring compliance with ISO/TS 15066 and other relevant industrial safety standards.

Restarting the process is quick and user-friendly, allowing for minimal downtime and improved workflow continuity on automated production lines.

This type of collaborative robotics solution is frequently used in applications such as automotive manufacturing, CNC machine tending, material removal, and applications where a balance of throughput, safety, and periodic manual intervention is essential.

Speed and Separation Cobots

Speed and Separation collaborative robots—often described as “fenceless automation solutions”—are designed for environments with fluctuating human presence and a need for unrestricted access to equipment. These cobots employ safety laser scanners, intelligent vision systems, and virtual safety zones to dynamically adjust the robot's operating speed based on a person's proximity.



When an operator enters the defined safety zone, the cobot slows operation, switching to safe-mode or stopping entirely as required. Once the area is clear, the cobot resumes its programmed speed, maximizing both workplace safety and operational efficiency. This technology is particularly promising for warehousing, logistics automation, and large-scale material

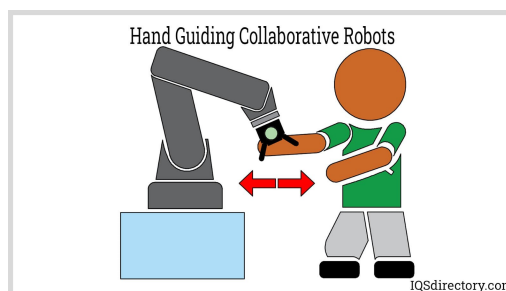
handling systems where flexible interaction and adaptive safety are vital.

Because speed and separation cobots offer exceptional adaptability, they can often replace traditional industrial robots in scenarios where safety, efficiency, and collaboration are equally important.

Hand Guiding Cobots

Hand guiding, or “lead-through teach,” collaborative robots feature safety-rated handles or end effectors that allow human operators to guide the robot arm manually for intuitive programming of complex paths, workflows, or pick-and-place operations. This user-friendly programming interface eliminates the need for advanced coding skills, enabling rapid deployment and task reconfiguration even by non-technical staff.

Hand-guided cobots are instrumental for applications that require frequent reprogramming, variation in production, or custom assembly operations—such as prototyping, low-volume manufacturing, or mobile workstations within dynamic environments.



Each of these four categories of collaborative robots offers unique benefits and levels of direct interaction between humans and robots. Cobots are designed to minimize the risk of accidents and injuries, leveraging the latest in robotics safety standards and smart automation features. As the rapidly evolving robotic automation industry continues to innovate,

new types and advanced models of cobots will emerge, further shaping the future of smart factories, Industry 4.0, and intelligent manufacturing.

Choosing the Right Collaborative Robot for Your Application

Selecting the best collaborative robot for your unique operation involves evaluating several key factors, including workplace safety requirements, payload demands, integration with existing automation systems, and the degree of required human-robot collaboration. Modern cobot manufacturers and leading robotics suppliers offer expert consultation, risk assessments, and customized automation solutions tailored to your needs. Investing in the right cobot technology can help streamline production, lower costs, improve workplace safety, and increase overall equipment effectiveness (OEE).

For organizations seeking to implement robotics in manufacturing, assembly, material handling, or laboratory environments, understanding the latest innovations in collaborative robotics is essential to maximizing ROI, productivity, and workforce satisfaction.

Leading Collaborative Robot Integrators and Suppliers

Dixon
Automati
c Tool,
Inc.

Invio
Automatio
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Advent
Design
Corporatio
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Isotec
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Del-Tron
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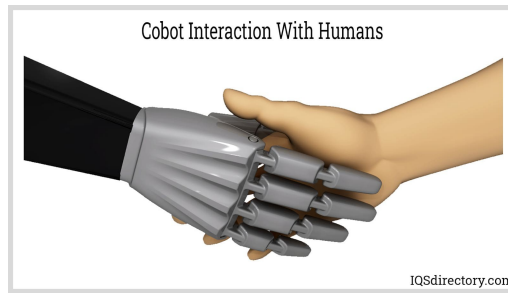
Chapter 3: What are the applications and benefits of collaborative robots (cobots)?

This chapter will explore the various applications and advantages of collaborative robots.

Impact of Collaborative Robots on Workers

Tasks such as packing goods, restocking inventory, and assembly line work are often repetitive and monotonous. These types of tasks can lead to Repetitive Strain Injury (RSI), a

condition resulting from the repeated performance of the same movements.



When robots take over repetitive tasks, human workers can focus on more complex and creative responsibilities, such as quality control or maintenance.

Assigning workers to solution-oriented and creative tasks not only enhances their job satisfaction but also fosters personal growth. This approach encourages flexibility and diversity, contributing to a more positive work environment. Such factors help workers find greater meaning in their roles, leading to increased engagement and productivity.

Applications of Collaborative Robots

Collaborative robots, equipped with various tools and intuitive software, are capable of handling a wide range of tasks. Their versatility allows them to integrate multiple functions into a single automated process. While theoretically, a cobot can be trained for numerous applications, certain tasks are commonly automated, as outlined below.

Sanding and Polishing

Tasks like sanding, polishing, or deburring require precise control. Thanks to built-in sensors, a collaborative robot can follow designated paths while applying a consistent force on the surface, ensuring uniform quality. This makes them ideal for processing wooden, stainless steel, or metal surfaces.

Screwing Jobs

Repetitive screwing tasks are well-suited for cobots. With the right tools and software, a cobot can be programmed to align objects with precision, apply consistent force, and maintain accuracy over long periods. This is challenging for humans but manageable for a cobot operating continuously throughout a workweek.

Pick-and-Place

Pick-and-place tasks involve picking up objects from bins and arranging them on conveyors. While this application is straightforward, it can become more complex with additional tools, such as double grippers or adjustable grippers for different materials and sizes.

Bin Picking

Bin picking is similar to pick-and-place but involves retrieving items from a non-standardized arrangement. Using vision systems and intelligent software, cobots can identify and pick items from bins one by one, and can also supply items to conveyor belts.

Machine Tending

Machine tending involves placing parts into machines like CNC lathes or bending equipment. By automating this task, operators are freed up to focus on more complex activities, enhancing safety and productivity while allowing for extended operational hours and increased output.

Cobots and Palletizing

Collaborative robots can significantly streamline the palletizing process. Using an electric vacuum gripper, a cobot can efficiently pick up boxes and place them onto pallets. Advances in technology have eliminated the need for external air supplies and hoses, simplifying integration and reducing costs.

Quality Testing and Inspection

Cobots are highly effective in performing quality control and inspection across various industries. They can handle tasks like placing items in testing equipment and sorting them based on visual inspections, thanks to advanced vision technology.

Gluing and Sealing

Gluing and sealing are ideal tasks for collaborative robots. By attaching a glue or sealant syringe to the cobot, it can follow a specified path and apply the adhesive consistently, ensuring uniform application.

Soldering and Welding

Soldering and welding require high precision, making collaborative robots an excellent choice for these tasks. Cobots offer consistent quality and precision in welding and soldering applications, surpassing human capabilities.

Different Industries for Collaborative Robots

Collaborative robots are highly adaptable and can be utilized across various industries. Their versatility allows them to be effectively integrated into many sectors, each with its unique requirements and challenges.

Food Industry

Automation in the food industry has increased due to factors such as higher demand and stricter safety regulations. While automating food handling presents challenges, advances in cobots and gripper technology have addressed these issues, enabling effective integration.

Plastics Industry

The plastics industry is characterized by low-volume, high-mix production, which presents challenges for automation due to

frequent changes in production processes. Collaborative robots, with their flexibility and ease of reprogramming, have adapted well to these requirements.

Packaging Industry

As the demand for customized packaging solutions grows, collaborative robots are increasingly used for tasks like packing products into boxes and stacking them on pallets. Cobots can handle physically repetitive tasks, improving both production efficiency and worker satisfaction.

Electronics Industry

In the electronics industry, where precision and careful handling of small components are crucial, collaborative robots equipped with advanced sensors are well-suited for tasks like quality control and testing. Their sensitivity and accuracy make them ideal for these applications.

Pharmaceutical Industry

The pharmaceutical industry benefits from automation due to its need for high accuracy and low error margins in large-scale production. Collaborative robots excel in maintaining consistent performance, reducing errors, and improving overall efficiency.

Automotive Industry

The automotive industry has long utilized industrial robots for assembly. With increasing demands for customization and small production runs, collaborative robots are now becoming valuable for various automotive applications.

Metal Industry

In the metal industry, where low-volume, high-mix production is common, collaborative robots are advantageous due to their adaptability. They can perform tasks such as soldering,

welding, and loading CNC machines efficiently, meeting diverse production needs.

Benefits of Collaborative Robots

Collaborative robots offer several advantages, including:

Compact – Cobots are compact and can be integrated into production environments without occupying significant space.

Installing and Programming – Cobots are easy to install and program. With user-friendly software for both desktops and smartphones, they can be up and running quickly.

Flexible – Cobots can be easily programmed to perform new tasks, making them versatile for various stages of the production process.

Mobile – Lightweight and easily movable, cobots can be placed on mobile workstations to perform different tasks in various locations within a facility.

Consistent and Precise – Cobots deliver consistent actions with precise force, ensuring uniform quality and accurate component placement.

Positive Impact on Employees – By handling dangerous or monotonous tasks, cobots allow employees to focus on more creative and meaningful work, enhancing job satisfaction.

Reduction of Costs of Production – The efficiency and increased production capabilities provided by cobots contribute to cost savings and improved profitability.

Drawbacks of Collaborative Robots

Collaborative robots may require a restricted working area to ensure worker safety. Initial setup costs can be high, and safety

measures might slow down operations. Cobots may need to stop when a person enters their workspace, impacting workflow. Additionally, some cobots are designed to slow down or shut down completely in the presence of humans, which could necessitate costly safety measures. Since they rely on human supervision, they cannot operate entirely independently.

To achieve effective integration, significant investment in safety and operational measures may be required. Collaborative robots' dependence on human oversight means they cannot function without continuous human support.

Safety Standards of Collaborative Robots

The RIA ISO/TS 15066 technical specifications and ISO 10218 safety standards provide guidelines on the safety performance and functionality of collaborative robots. Published in 2016, the RIA TR15.606-2016 Cobots document outlines safety

requirements for robot systems and collaborative robots, reflecting the U.S. adoption of ISO/TS 15066. Under TS 15066, collaborative robots' speed and force limits are configured based on human contact areas, application data, and workspace hazards.

The RIA TR15.806-2018 Guidance on Testing Force and Pressure in Collaborative Robot Applications offers guidelines for measuring device conditions, test measurements, and testing methods. It includes metrics and methods for evaluating forces and pressures associated with both transient and quasi-static contact in collaborative applications.

Conclusion

Cobots have leveled the playing ground for small to medium-sized industries. For years, large industries have had the benefits of automation by utilizing traditional industrial robots.